Doctoral Research Dossier

Civic auditing and reproducibility in LLM-based multi-agent systems

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Project repository on GitHub

Purpose and evidence status

This dossier organizes the supplied MARPA-v1 brief into seven thematic blocks. It preserves researcher-reported figures while separating them from independently verified results, proposed procedures and missing evidence. It is a documentation and presentation outline, not a record of a completed replication.

No MARPA source code, dataset manifest, prediction file or experimental log accompanied the brief. The existing repository reviewed earlier documents a thesis protocol and a DelphiAgent audit, rather than a validated MARPA experiment. Claims about licences, funding, institutional partnerships, dataset deposits and contact details therefore require confirmation before being stated as established facts.

Reading guide

Block	Content
1	Problem formulation and fair benchmarking
2	Leakage prevention
3	Uncertainty and statistical comparisons
4	Ablation analysis
5	Nine-section Model Card
6	Seven-section Datasheet
7	Governance mapping and documentation scorecard

The seven blocks follow the attached dossier, not the numbering of the existing GitHub folders. They support an oral presentation but do not constitute seven timed five-minute transcripts.

1 Problem formulation and fair benchmarking

The civic auditing problem

MARPA-v1 is specified as a research system for reviewing Peruvian state responses to access-to-information requests under Law 27806. Target issues include possible administrative silence, partial disclosure and missing procurement annexes. Institutional level and document quality are relevant evaluation factors, rather than evidence by themselves of non-compliance.

An audit must connect a request to its response and supporting records. A missing retrieved document does not establish that an institution unlawfully withheld it. Delivery dates, extensions, applicable exceptions and the completeness of the evidence package require human review.

The methodological contribution

The intended contribution is a modular multi-agent architecture that makes extraction and verification inspectable. The related thesis protocol adopts Design Science: the artifact must be evaluated against alternatives, with explicit measures of utility, error and resource use.

Ferrari Dacrema, Cremonesi and Jannach [1] motivate caution about weak baselines in recommender-system research. The transferable lesson here is to compare against competent, tuned alternatives. This does not require pretending that different architectures consume identical resources: tuning opportunities, evidence access and compute budgets should be transparent, with controlled comparisons where feasible.

Reported comparison

Configuration	Accuracy	Full-system advantage
Untuned logistic regression	0.692	0.172 = 17.2 percentage points
Tuned logistic regression	0.814	0.050 = 5.0 percentage points
MARPA-v1	0.864	Reference

The draft reports C = 0.001 for the untuned baseline and a 12-value search between 10^-3 and 10^2 using five-fold cross-validation for the tuned baseline. The exact grid, scoring objective, feature representation, solver and fold assignments were not supplied. These details must be recorded rather than inferred from the endpoints.

The reported paired p-value for MARPA versus tuned logistic regression is 0.012. It is transcribed from the brief, not recomputed. Without paired outputs and a named analysis, the gain is a descriptive reported difference rather than an independently confirmed superiority claim.

Evidence required

Release the baseline configurations, development-only tuning results, common test manifest, per-case predictions, failure policy and compute measurements. A weak baseline can remain as a diagnostic comparison, but it should not be the main basis for claiming progress.

2 Structural safeguards against data leakage

Why leakage matters

Leakage occurs when information unavailable at prediction time influences model development. It may inflate scores without causing a software error. Cawley and Talbot [2] discuss selection bias in performance evaluation; scikit-learn documentation [6] recommends learning preprocessing transformations on training data only.

The laboratory example in the supplied draft

The draft describes an experiment with 5,000 Gaussian noise features: feature selection before cross-validation allegedly produced accuracy above 73%, while a pipeline-based approach produced 57%. No code, sample size, label-generation method, split or uncertainty estimate was supplied. These values are retained as an unverified laboratory example, not as a reproduced finding or a numerical result established by reference [2].

An observed 57% accuracy cannot automatically be called chance performance. The expected baseline depends on the label distribution and decision rule. Even with balanced binary labels, sampling uncertainty is needed to determine whether the result is compatible with a 50% expectation.

Declared MARPA safeguard

The supplied protocol places preprocessing and scaling inside scikit-learn Pipelines, with learned transformations fitted only on the training portion of each cross-validation fold. This includes learned feature selection, vocabulary construction, imputation and scaling. A fixed text-normalization rule may not learn parameters, but its configuration must still be documented.

Additional safeguards required

Split at case level. Keep related requests, annexes, appeals and revised responses together. Check exact and near duplicates before assigning development and test partitions.

Protect the sealed test set. Do not use test labels or examples to select prompts, tune retrieval, choose thresholds or calibrate probabilities.

Version the RAG evidence. Record corpus snapshots and effective dates. Exclude test annotations and other answer-bearing material from retrieval indexes. Avoid using later evidence in an evaluation of an earlier decision.

Control OCR and caches. Select learned OCR corrections on development data. Record OCR versions and parameters, and include dataset and transformation versions in cache keys.

Pipelines reduce fitting-related leakage; they do not by themselves prove that all these other routes have been controlled. The claim that safeguards are enforced throughout the repository remains pending code and log evidence.

3 Statistical rigor and multiple comparisons

Repeated runs and distinct sources of uncertainty

The brief reports five seeds and mean ± standard deviation. Repeated runs help describe variability, but the exact seeds and what they control must be recorded. Five runs on the same 300 cases are not 1,500 independent test cases. Run-to-run variability and uncertainty over the target population are different quantities.

API temperature T = 0.1 does not guarantee deterministic outputs. Exact model identifiers, decoding settings, request dates, retry policies and response records are needed to assess repeatability. No blanket claim of compliance with a conference checklist is made here.

Proposed inference toolkit

Paired t-test. A test on seed-level differences requires matched runs and justified assumptions about those differences. With five pairs, assumptions are difficult to assess. It tests a mean difference; it does not establish arbitrary distributional differences.

Exact McNemar test. For two fixed classifiers evaluated on the same cases, the test uses discordant correct/incorrect predictions. It addresses a difference in error rates, not directly macro-F1 or calibration. Independent sampling units and repeated-case dependencies must be considered [3].

Paired bootstrap. The draft proposes 2,000 resamples. Resample the same case indices for both systems, or groups of related cases where necessary, and state the interval method. No bootstrap intervals were supplied, so none are presented as computed results.

Multiplicity correction

Bonferroni controls the family-wise error rate, not the false discovery rate. For k independent tests under their null hypotheses, 1 - (1 - alpha)^k gives the probability of at least one false rejection. The independence qualification is essential; Bonferroni itself does not require independence.

As a clearly labelled sensitivity calculation, if the six non-reference comparisons in Block 4 form one family, alpha = 0.05 gives a Bonferroni threshold of 0.00833. On the supplied rounded p-values, 0.008, 0.002 and <0.001 fall below that threshold; 0.012, 0.021 and 0.142 do not. This arithmetic is not a reanalysis of the experiments, and the comparison family was not documented as preregistered.

Evidence required

Provide per-run and per-case outputs, the test used for each claim, the pairing unit, confidence intervals, effect sizes, multiplicity policy and a dated preregistration if one exists. The draft’s assertion that these analyses were implemented has been changed to a proposed analysis plan pending evidence.

4 Ablation analysis

Each component ablation should disable one defined component while retaining the same test cases and controlled settings. This estimates sensitivity to that intervention under the tested configuration. It does not automatically establish real-world causality or additive contributions because modules can interact.

Researcher-reported figures

Configuration	Accuracy mean ± SD	Macro-F1	Delta accuracy	Raw p
Full MARPA-v1	0.864 ± 0.012	0.851	Reference	-
No deterministic citation verification	0.803 ± 0.018	0.789	-0.061	0.008
No arbiter; majority vote	0.821 ± 0.015	0.806	-0.043	0.021
No structured OCR preprocessing	0.772 ± 0.024	0.755	-0.092	0.002
No generic legal context injection	0.849 ± 0.014	0.835	-0.015	0.142
Tuned logistic regression	0.814 ± 0.011	0.801	-0.050	0.012
Untuned baseline	0.692 ± 0.029	0.668	-0.172	<0.001

All entries originate from the attached brief. Accuracy differences were checked against the full-system mean. Macro-F1 values for the non-reference rows appear in this attachment but were not included in the earlier technical specification. They remain reported values pending prediction-level verification. Macro-F1 standard deviations for these rows were not supplied.

Interpretation

Removing OCR preprocessing is associated with the largest reported accuracy reduction among component ablations, followed by removing deterministic citation verification. These reductions should not be added together or described as accounting for most performance without an interaction analysis.

Removing generic legal context produces a reported 0.015 reduction with $p = 0.142$. The appropriate interpretation is that the supplied test does not establish a difference at a nominal 0.05 threshold. It is not proof of equivalence, negligible practical effect, or pure noise. An equivalence claim would require a justified margin and an appropriate analysis.

The small p-value for the untuned baseline is not itself misleading. The problem would be using an inadequately tuned comparison as the main evidence of architectural superiority. Statistical significance cannot repair an unfair experimental design.

Configuration questions to resolve

Specify what remains when OCR preprocessing is removed, which predictions form the majority vote, and how generic legal injection differs from legal retrieval within the verifier. Pin the backbone across component ablations. Publish raw paired outputs before using the table as validated experimental evidence.

5 Model Card

System-level card organized according to the nine sections in Mitchell et al. [4]. Status: researcher-supplied specification and results, with verification gaps stated explicitly.

1 Model details

Multi-Agent RPA Auditor, MARPA-v1. Author: Mayte Ciriaco Ruiz, UNMSM. The specified DAG contains a civic request extractor/decomposer, an evidence verifier using SUNARP, OSCE and Law 27806 sources, and a compliance arbiter with probability calibration. RAG retrieval is distinct from deterministic rule checking.

Reported backbones: local Llama-3-70B-Instruct and API-based Claude-3.5-Sonnet/GPT-4o at T = 0.1. Exact snapshots, checkpoint hashes, serving configuration and per-backbone results are unknown. MIT and CC BY-NC 4.0 licences are asserted in the draft but not verified; confirm ownership, upstream terms and licence files before publication.

2 Intended use

Assist investigative journalists, civic auditors and transparency officers in extracting and checking public-information responses. Human review is required before legal or public allegations. Out of scope: autonomous sanctions, binding legal rulings, requester profiling, restricted sensitive records and abusive mass-request campaigns.

3 Factors

Intended language: Peruvian administrative Spanish. Relevant factors include institutional level, document type, scan quality, completeness and retrieval coverage. Performance in other languages or populations is not established.

4 Metrics

Reported: accuracy, macro-F1 and Brier score. Proposed supplementary measures include precision/recall, negative log-likelihood and expert agreement. Their definitions and outputs are missing. Weighted kappa requires an appropriate ordered label scheme; nominal disclosure categories should not receive arbitrary ordinal weights. The proposed threshold 0.5 needs a task-specific decision rule, especially if the task is multiclass.

5 Evaluation data

Reported sealed test set: 300 real administrative case files. The draft names full disclosure, partial disclosure, formal denial and silence as categories, but provides no counts or label guide. Split hashes, overlap checks and annotation evidence remain missing.

6 Training data

Reported development/train set: 1,000 de-identified cases. The attachment attributes them to TTAIP and OSCE materials from 2021-2025 and describes Lima-heavy coverage; these provenance and distribution claims need a manifest. Use for development does not establish that backbones were fine-tuned.

5 Model Card continued

7 Quantitative analyses

Measure	Reported result
Accuracy	0.864 ± 0.012
Macro-F1	0.851 ± 0.015
Brier score	0.118; definition and normalization unspecified

Slice	Precision	Recall	F1
Native PDFs	0.912	0.895	0.903
Scanned PDFs, low/medium OCR	0.798	0.741	0.768
Central government	Not supplied	Not supplied	0.884
Regional/municipal government	Not supplied	Not supplied	0.782

Slice counts, confidence intervals and averaging conventions are absent. The precision and recall pairs are numerically consistent with the displayed F1 values to rounding under a single harmonic-mean calculation; this does not validate the underlying observations or establish a macro-averaging method.

A Brier score of 0.118 alone does not establish calibrated uncertainty. The class structure, formula, reference predictor and reliability analysis are needed. Aggregate metrics cannot be assigned to a particular backbone from the supplied information.

8 Ethical considerations

Incorrect flags can harm reputations, while missed omissions can weaken accountability. Poor scans may create apparent omissions. Proposed safeguards include an insufficient-evidence outcome, human escalation for conflicting evidence, links to source folios and dated legal provisions, and protection of personal data.

The brief describes automated redaction and complete traceability as implemented. Without code or audit logs, this dossier records them as declared controls awaiting verification. Human review should permit corrections and overrides rather than merely approve the automated conclusion.

9 Caveats and recommendations

The asserted degradation below 200 DPI is unverified and should be tested rather than used as an established cutoff. The legal corpus must track effective versions, including the consolidated transparency law identified in reference [7]. API reproducibility, OCR robustness, institutional transfer and evidence completeness remain evaluation requirements.

Use MARPA only as an analytical aid during appeals or journalistic investigations. Before deployment, supply independent validation, release criteria, source-linked error analysis, calibrated escalation rules and a documented privacy assessment. These recommendations do not amount to legal certification.

6 Datasheet for Datasets

Peru-FOIA-Audit-Bench v1.0. Seven lifecycle sections following Gebru et al. [5]. This records the supplied dataset description, not a verified dataset release.

1 Motivation

The intended purpose is a Spanish-language benchmark for extraction and evidence-based audit of Peruvian public-information responses. Mayte Ciriaco Ruiz is the named researcher. The attachment's claims of a UNMSM laboratory partnership with El Comercio, academic funding and absence of political funding were not supported by documentation; institutional authorization and funding records are pending.

2 Composition

Reported total: 1,300 administrative case files, with 1,000 for development/training and 300 for sealed testing. A case file may contain several PDFs. The draft describes requests, notices, annexes and appeal decisions, but field completeness and class counts are unknown. Do not call every dossier complete when missing material is itself a target condition.

3 Collection process

The attachment reports acquisition through TTAIP/Ministry of Justice sources and reporting-related information requests, covering 2021-2025 and closing in February 2026. Source URLs, access basis, collection logs, sampling rules and closure evidence were not supplied. Proposed exclusions include legally protected or classified material; an ethics-review decision has not been established.

4 Preprocessing cleaning and labeling

Reported procedures include OCR, conversion of tables to JSON/HTML, anonymization, legal-expert double annotation and SHA-256 hashes for originals. The tools, label guide, annotator agreement, disagreement procedure and hash manifest are missing. The earlier thesis protocol proposed at least three annotators and a fourth adjudicator; any move to double annotation requires a documented protocol amendment.

Fit learned transformations within development training folds. Audit identifiers in text, images, hidden layers and metadata. Distinguish anonymization from reversible pseudonymization. Law 29733 and its regulation frame the privacy assessment [8]; a statement of redaction alone does not prove compliance.

5 Uses

Intended uses: civic technology, document extraction and verification research. Prohibited uses: political profiling, blacklisting requesters and autonomous punitive decisions. Do not infer national non-compliance rates from a sample with an undocumented sampling frame.

6 Distribution

Zenodo/Hugging Face hosting, a dataset DOI and CC BY-NC 4.0 licensing are asserted in the draft but no deposit URL or identifier was supplied. Distribution status is unverified. Public-document provenance does not by itself authorize redistribution of all personal data or third-party content.

7 Maintenance

A biannual review is proposed. The institutional maintenance team and email address in the attachment are unverified and have been omitted as operational contacts. Confirm a responsible custodian, update policy, correction/withdrawal process and retention schedule. Preserve v1.0 test membership and issue a new version for material corrections.

7 Governance and documentation scorecard

National framework and international comparison

The design aims to support transparency, explainability and human oversight in relation to Peru's Law 31814 and its regulation, Supreme Decree 115-2025-PCM [9]. Personal-data processing requires a documented assessment under Law 29733 and its regulation [8]. No legal compliance audit or deployment approval was supplied.

EU Regulation 2024/1689 Annex IV [10] is used as a technical-documentation reference. EU territorial applicability and high-risk classification require a separate assessment. A Peruvian research prototype is not automatically an EU high-risk system.

Annex IV area	Available description	Outstanding evidence
1 System description	Purpose and architecture	Versions, interfaces and user instructions
2 Development	Proposed data and pipeline	Design records and validation logs
3 Monitoring/control	Human review and slices	Implemented oversight controls
4 Metrics	Reported performance	Definitions and checked predictions
5 Risk management	Harms identified	Risk register and mitigation tests
6 Lifecycle changes	Versioning recommended	Change history and impact reviews
7 Standards	References named	Applicable standards and evidence
8 Declaration	None claimed	Only if required and substantiated
9 Post-market monitoring	Review proposed	Formal monitoring and incident plan

Documentation scorecard from the supplied brief

Item	Coverage in this dossier
Model Card has nine sections	Present; Blocks 5 and 5 continued
Datasheet has seven sections	Present; Block 6
Misuse and out-of-scope uses stated	Present; Blocks 5 and 6
Performance includes variation	Reported mean ± SD; raw runs pending
Metrics disaggregated	Reported slices; subgroup counts pending
Dataset prohibited uses stated	Present; Block 6
Collection process described	Reported process; provenance pending
Unknowns and gaps identified	Explicit throughout

Documentation coverage: 8/8 categories addressed. This is not instructor approval, independent reproducibility certification or proof of compliance. The exact Session 7 Exercise 6 rubric was not supplied for authentication. "Submission ready" has therefore been replaced by a statement of coverage and evidence status.

Release requirements and references

Before presenting the dossier as completed experimental work

Reconcile this specification with the repository protocol: the earlier protocol sets a future minimum of 500 documents, while this brief describes a completed 1,300-case benchmark. Document the dates, sampling expansion and protocol changes. Keep the DelphiAgent audit distinct from MARPA evaluation.

Provide the MARPA commit, environment lock, exact seed list, model snapshots, prompt versions, RAG snapshots, dataset/split manifests, annotation records, per-case predictions and statistical scripts. Confirm licences, institutional roles and distribution rights. No environment version or seed sequence has been invented here.

The attachment lists model_card.md, datasheet.md, annex_iv_crosswalk.md, REPRODUCIBILITY_REPORT.md and two figures as existing bundle contents. Their existence was not verified. They are recommended future artifacts; this PDF does not claim that a ZIP, compute-matched sweep or ablation error-bar plot has been generated.

Editorial review and AI-use declaration

This version removes copied interface labels and unresolved citation tokens. It corrects percentage-point interpretation, the claim that 57% proves chance performance, the confusion between family-wise error and false discovery rates, and the interpretation of a non-significant result as noise. Supplied numerical results are preserved with explicit provenance.

An OpenAI AI assistant reviewed and organized the supplied text and generated this PDF using Python and ReportLab. It did not execute MARPA experiments, validate the private corpus or certify compliance. The author should verify supporting evidence before representing the reported results as established findings.

References

[1] Ferrari Dacrema, M., Cremonesi, P., & Jannach, D. (2019). Are We Really Making Much Progress? A Worrying Analysis of Recent Neural Recommendation Approaches. RecSys 2019, 101-109. Source.

[2] Cawley, G. C., & Talbot, N. L. C. (2010). On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. JMLR, 11, 2079-2107. Source.

[3] Dietterich, T. G. (1998). Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms. Neural Computation, 10(7), 1895-1923. Source.

[4] Mitchell, M., et al. (2019). Model Cards for Model Reporting. FAT* 2019, 220-229. Source.

[5] Gebru, T., et al. (2021). Datasheets for Datasets. Communications of the ACM, 64(12), 86-92. Source.

[6] scikit-learn. Common pitfalls and recommended practices, data leakage. Source.

[7] Peru. Supreme Decree 021-2019-JUS. Consolidated text of Law 27806. Source.

[8] Peru. Supreme Decree 016-2024-JUS. Regulation of Law 29733. Source.

[9] Peru. Supreme Decree 115-2025-PCM. Regulation of Law 31814. Source.

[10] European Parliament and Council. Regulation (EU) 2024/1689, Annex IV. Source.